Non-Invasive Anemia Detection

Estimating haemoglobin from photographs of the palpebral conjunctiva

Progress report · 13 August 2026

1. Summary

This period was spent building the image-processing and learning pipeline end to end: data loading and clinical labelling, conjunctiva extraction, quality control, the regression scaffold, an evaluation methodology, and a serving API. The guiding constraints are an eventually deployed screening app and small clinical cohorts — both punish silent failure, so the pipeline is engineered to produce numbers that can be defended: patient-grouped cross-validation, out-of-fold reporting only, and every result compared against a predict-the-mean baseline.

A real labelled cohort was obtained: 52 photographs of 26 children, both eyes, with laboratory haemoglobin and a full blood count. Screening it produced two findings. The pipeline processes real photographs without error and 94% pass quality control, but a plain colour-threshold extraction — which scores almost perfectly on synthetic images — proved **unreliable on real tissue**; a refined extractor was then built against these captures and now produces clean masks across all 52. Separately, the per-patient WHO threshold was validated against real laboratory values: a fixed single cutoff would have missed 15% of the anemic children in this cohort. Both are reported in Section 4.

The regression model has **not** yet been trained on real patient photographs, so no accuracy figure in this report is a clinical result.

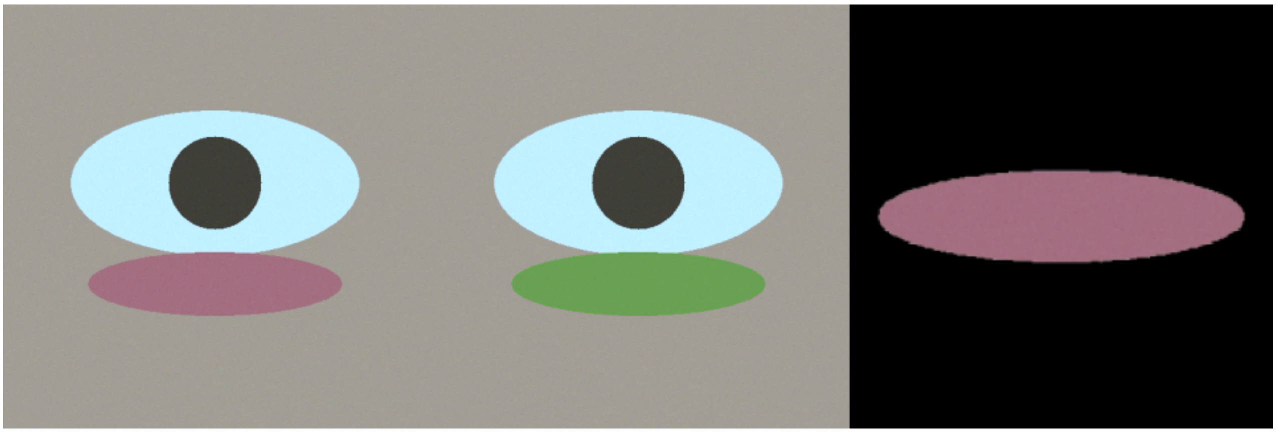
2. Pipeline design

The pipeline is a package of focused modules. The decision that matters most architecturally is that the inference path shares its preprocessing code with training but imports none of the training stack — so the deployed server cannot drift from the conditions the model was trained under, and does not load training machinery to answer a request.

Area	Design
Extraction	Built around a redness prior in CIELAB refined by a seeded grabCut; all backends share one interface so a trained model swaps in without touching other stages
Quality control	Enforced, not merely recorded: failing captures are dropped from training; the API returns a retake prompt rather than a number
Clinical labelling	WHO thresholds applied per patient from age and sex
Evaluation	Patient-grouped, haemoglobin-stratified cross-validation; out-of-fold predictions only; always reported against a predict-the-mean baseline
Checkpoints	Self-describing bundle carrying weights, standardisation constants, preprocessing settings and metrics
Performance	Preprocessing cached to disk, so segmentation runs once rather than once per epoch
Data handling	Adapters for both public datasets behind a common record type; adding the hospital data requires one adapter
Deployment	FastAPI service loading one model at startup
Testing	19 automated tests, plus synthetic eye phantoms allowing development without data

3. Choosing the extraction target: a controlled comparison

Two candidate colour rules were scored against known ground truth on synthetic phantoms. A brightness-based mask (bright, low-chroma pixels) scores a Dice coefficient of **0.000** — it selects the sclera and has no overlap with the conjunctiva. The redness-based extractor scores **0.999**. This is what fixed redness, not brightness, as the extraction target.



Left: input capture. Centre: region identified by the redness-based extractor, tinted green. Right: the square, undistorted crop passed to the regression network.

> **These are synthetic phantoms with flat colours and artificially clean separation.** A near-perfect score is expected and carries no clinical meaning. What the comparison establishes is that brightness-based masking targets the wrong tissue and redness-based extraction targets the right one. Section 4 shows what happens on real photographs.

4. Evaluation on real photographs

A local cohort of 26 children was obtained: both eyes photographed, giving 52 close-up captures of the everted lower eyelid, accompanied by laboratory haemoglobin, date of birth, gender, and a full blood count. The captures are well framed and in focus — tighter on the conjunctiva than typical field photography.

4.1 Cohort

Property	Value
Patients / images	26 / 52 (both eyes, exactly two captures each)
Haemoglobin	8.1 – 14.3 g/dL (mean 11.2, sd 1.37)
Anemic by WHO threshold	13 of 26 (50%)
Severe (Hb < 9.0)	3 patients
Age	3.4 – 16.7 years
Additional labels	Height, weight, socio-economic status, HCT, RBC, MCV, MCH, MCHC, RDW, platelets, MPV, TLC

Two properties make this cohort more useful than its size suggests. The 50/50 anemic split is unusually balanced — most cohorts skew heavily normal, which is what makes a predict-the-mean baseline hard to beat. And because both eyes of one child are photographed, the two captures are highly correlated: the loader assigns them a shared patient identifier so the grouped splitter cannot separate them; under a naive image-level split this dataset would produce substantially inflated results.

4.2 The pipeline runs on real data

Measure	Result
Images processed without error	52 of 52
Passed quality control	49 of 52 (94%)
Focus (Laplacian variance)	median 85.3, range 25.4 – 294.2

Measure	Result
Mask coverage of frame	median 0.211, range 0.041 – 0.449
Rejections	3, all blur (25.4, 34.2, 34.9 against a threshold of 35.0)

Two of the three rejections sit marginally below the threshold on captures that look usable to the eye. The blur threshold was chosen on synthetic images and appears slightly too aggressive for real photographs; it should be recalibrated once ground-truth masks make the downstream effect measurable.

4.3 Extraction: failure found, then fixed

Quality-control statistics say whether a capture is usable, not whether the segmentation found the right structure. Visual inspection of the overlays answers that, and the answer is unfavourable: on a substantial fraction of the 52 real captures the colour prior bleeds onto the eyelashes and lower-lid skin, and occasionally onto the sclera. On several it extends well below the lash line onto the cheek.

The cause is straightforward: eyelashes and peri-orbital skin are reddish-brown enough to pass a redness test, and neither is dark enough in absolute terms to be removed by a lightness gate.

A refined extractor was then built against these captures. It seeds a mask-initialised grabCut with what actually distinguishes the tissue: conjunctiva is smooth where the lash zone is high-frequency (a local-texture gate), sclera is bright and desaturated, iris and pupil are dark, and specular highlights sit on the wet tissue itself and are kept rather than discarded. Re-screened over all 52 captures, the refined extractor produces clean, tissue-hugging masks on visual review, and the median mask area halves — consistent with the bleed being removed rather than the tissue. The residual failure mode is a barely-everted lid, which is a capture problem rather than a segmentation one. The figure below shows the plain threshold (top) against the refined extractor (bottom) on the same captures.

Four real captures. Top row: the plain colour threshold, which bleeds onto lashes and skin. Bottom row: the refined seeded-grabCut extractor on the same images.

> **A method scoring 0.999 on synthetic images was visibly unreliable on real tissue** — exactly the gap synthetic validation cannot reveal. The failure was characterised, a refined extractor was built against the real captures, and visual review across all 52 now shows clean masks. What remains missing is a *measured* Dice on real photographs, which requires the Eyes-defy-anemia ground-truth masks; until then the learned segmenter remains the primary long-term path.

4.4 The per-patient threshold validated on real data

A single fixed anemia cutoff (commonly 11.0 g/dL) is a tempting simplification, and this cohort measures exactly what it would cost, because its ages span three WHO threshold bands: 11.0 below five years, 11.5 from five to eleven, and 12.0 from twelve to fourteen. Twenty of the twenty-six children fall in the middle band.

Patient	Hb (g/dL)	Age	WHO threshold	Correct	Fixed 11.0 rule
10	11.0	11.1	11.5	anemic	**normal**
14	11.4	6.6	11.5	anemic	**normal**
16	11.3	8.4	11.5	anemic	**normal**
21	11.4	7.2	11.5	anemic	**normal**

> **A fixed 11.0 rule misclassifies 4 of 26 patients (15%), and every one is a false negative — an anemic child labelled normal.** The errors cluster just below 11.5 because most of this cohort sits in the five-to-eleven band, where the correct cutoff is half a gram above the fixed value. This error occurs before the model makes a single prediction, so no amount of regression accuracy could recover it — which is why the pipeline predicts a continuous Hb and applies the threshold per patient.

5. Verification status

5.1 Verified

- All pipeline stages execute end to end; 19 of 19 automated tests pass.
- Cross-validated training completes on synthetic data: MAE 0.664 ± 0.096 g/dL, R^2 0.880, 1.45 g/dL better than a predict-the-mean baseline.
- A saved checkpoint carries everything needed to serve it, loads, and produces a haemoglobin estimate.
- Quality control correctly rejects blurred captures and returns a retake prompt.
- Clinical thresholds resolve correctly across age and sex.
- The pipeline processes real photographs end to end: 52 of 52 without error, 94% passing quality control.
- The per-patient WHO threshold is validated against real laboratory values: a fixed single cutoff misclassifies 15% of this cohort, all as false negatives (Section 4.4).

5.2 Measured and found wanting

- The plain colour threshold is unreliable on real tissue, bleeding onto eyelashes and lid skin. A refined seeded-grabCut extractor built against the same captures now produces clean masks on visual review across all 52 (Section 4.3); a measured Dice still awaits ground-truth masks.
- The blur threshold looks slightly too aggressive for real captures: two of three rejections were marginal on images that appear usable.

5.3 Not yet established

- Any *accuracy* on real photographs. The local cohort is labelled and could in principle supply one, but 26 patients is far too few for a defensible confidence interval, and segmentation on these images is known to be unreliable. Training on it is worthwhile only after the segmenter works.
- The segmentation model has not been trained — this requires the ground-truth masks.
- The CP-AnemiC loader has not been checked against a real download.
- No clinical validation of any kind has been performed.

6. Data

Two public datasets were identified and assessed, and a small labelled cohort is held locally. The professor has arranged hospital data, which will be the primary source; these sources allow development and benchmarking in the meantime.

Source	Images	Hb	Masks	Role
Eyes-defy-anemia	218	Yes	Yes	Only source of pixel-level ground truth; trains the segmenter
CP-AnemiC	710	Yes	No	Additional volume for the regressor
Local cohort	52	Yes	No	26 children, both eyes, with full blood count. Balanced 50/50 anemic. Too small to train on alone

Two consequences follow. First, roughly 900 images is a small corpus: it constrains model size, makes cross-validation necessary rather than optional, and means confidence intervals belong on every reported number. Second, the two sets differ in age, ethnicity, camera and lighting, so a model trained on one and evaluated on the other measures domain transfer as much as pallor. Results should be reported per source as well as pooled.

The local cohort carries a full blood count alongside haemoglobin. RDW and MCV distinguish iron-deficiency anemia from other types, which raises a possible extension: predicting which anemia rather than only whether. That is beyond current scope but worth recording. Provenance and licensing are unconfirmed, which matters if any figure derived from these images appears in a published write-up.

7. Risks

Risk	Mitigation
Small dataset produces a model that does not beat the mean baseline	Baseline comparison reported by construction, so this cannot go unnoticed
Colour heuristic proves inadequate on real photographs	Expected; the Mask2Former segmenter is the primary path and the heuristic only a fallback
Hospital data arrives in an unanticipated format	Loaders are isolated behind a common record type; an inspection command verifies parsing before training
Results overstated in the write-up	Grouped splits, out-of-fold reporting and baseline comparison are built into the pipeline rather than left to discipline

8. Next steps

#	Step	Outcome
1	Download both datasets and verify parsed counts against published figures	Confidence the loaders are correct
2	Train the segmentation model on Eyes-defy-anemia	First trained component, and the first measured Dice for the refined extractor
3	Benchmark all segmentation backends against ground truth	**First real measurement in the project**; quantifies the rework
4	Train the regressor; report per-source and pooled with confidence intervals	First meaningful accuracy estimate
5	Recalibrate the blur threshold against real captures	Fewer spurious retake prompts
6	Pool the local cohort with the public datasets once segmentation is reliable	26 balanced, well-labelled patients added to training
7	Build the application around the existing API	Working demonstrator

Assessment. The pipeline is in a state where real data can be introduced and produce trustworthy numbers, and it has been exercised on real photographs rather than synthetic ones. Extraction — this period's focus — was taken from a failing colour threshold to clean masks across the full real cohort, with the remaining step being a measured Dice against ground-truth masks. The scientific question, whether conjunctival pallor predicts haemoglobin accurately enough in this cohort to be useful, remains open and is what the next phase must answer.